

The Transcendental Volume Function

Kähler MMP

Yi Li

The aim of this note is to introduce the transcendental volume.

Contents

1	Definition of volumes	1
2	Fujita's approximation	3
3	Basic properties of volume functions	3
3.1	For $(\mathbb{R})$ -divisors	3
3.2	For cohomology classes	5
4	Upper semi-continuity of volumes	5
5	Deformation invariance of volume of adjoint divisors (classes)	6
6	Volume in divisorial Zariski decomposition	6
7	Volume in the minimal model program	7
8	Jiao's proof of upper semi-continuity of volume function	7

1 Definition of volumes

Definition 1.1 (Volume of a line bundle, [Laz04, Definition 2.2.31]). Let X be an irreducible projective variety of dimension n , and let L be a line bundle on X . The volume of L is defined to be the non-negative real number

$$\mathrm{vol}(L) = \mathrm{vol}_X(L) = \limsup_{m \rightarrow \infty} \frac{h^0(X, L^{\otimes m})}{m^n/n!}.$$

The volume $\mathrm{vol}(D) = \mathrm{vol}_X(D)$ of a Cartier divisor D is defined similarly, or by passing to $\mathcal{O}_X(D)$.

Definition 1.2 (Volume of a $\mathbb{R}$ -divisor, [FKL16, p. 8]). If X is a normal projective variety, D is an $\mathbb{R}$ -divisor and $n = \dim X$, then we define the volume of D by

$$\mathrm{vol}(X, D) = \limsup_{m \rightarrow \infty} \frac{n! h^0(X, mD)}{m^n}$$

where $h^0(X, mD) := h^0(X, \lfloor mD \rfloor)$.

Definition 1.3 (Volume of cohomology classes, [Bou02]). Let X be a compact Kähler n -fold. We define the volume of a cohomology class $\alpha \in H^{1,1}(X, \mathbb{R})$ by

$$\mathrm{vol}(\alpha) := \sup_{T \in \alpha} \int_X T_{ac}^n = \sup_{T \in \alpha} \int_{X \setminus \mathrm{Sing}(T)} T^n$$

for T ranging over closed positive $(1, 1)$ -currents in α , in case α is pseudo-effective. If it is not, we set $\mathrm{vol}(\alpha) = 0$.

Remark 1.4 (Lebesgue's decomposition and absolutely continuous part). A current T on an n -fold X can be seen as a form with distribution coefficients. When T is positive, the distributions in question are positive measures which admit a Lebesgue decomposition into an absolutely continuous part (with respect to the Lebesgue measure on X) and a singular part.

We therefore get a decomposition of T itself into an absolutely continuous part T_{ac} and a singular part T_{sing} . The decomposition

$$T = T_{ac} + T_{sing}.$$

The reason to take the absolutely continuous part is that the n -fold product of an arbitrary closed positive current T is not well defined (it requires a locally bounded potential). Once we assume T has locally bounded potentials, the product T_{ac}^n is well defined via Bedford–Taylor theory.

However, the decomposition does not behave well at the cohomological level. As a closed positive current T , the absolutely continuous part T_{ac} need not be closed.

Remark 1.5 (Volume on normal varieties). On a compact normal Kähler variety, we can first take a resolution $\pi : X' \rightarrow X$, and define the volume of a class $\alpha \in H_{BC}^{1,1}(X, \mathbb{R})$ to be

$$\mathrm{vol}(\alpha) = \sup_{[T] = \pi^* \alpha} \int_{X'} T_{ac}^n.$$

It is well defined, since the volume function for a smooth variety is a birational invariant (see Theorem 3.4).

One can show that the analytically defined volume coincides with the algebraically defined volume.

Theorem 1.6 ([Bou02, Theorem 1.2]). *Let L be a pseudoeffective line bundle on a compact Kähler manifold X . Then the volume of L satisfies*

$$\mathrm{vol}(L) = \max_T \int_X T_{ac}^n$$

for T ranging among the closed positive $(1, 1)$ -currents in the cohomology class $c_1(L)$.

2 Fujita's approximation

One of the central properties that the volume function satisfies is the Fujita approximation. Both big line bundles and big cohomology classes satisfy this property.

Theorem 2.1 (Fujita approximation (for big line bundle on projective variety)). *Let L be a big line bundle on a projective manifold X . Then, for every $\varepsilon > 0$, there exist a modification $\mu : \tilde{X} \rightarrow X$, an ample $\mathbb{Q}$ -line bundle A , and an effective $\mathbb{Q}$ -divisor E on $\tilde{X}$ (all depending on ε) such that:*

- (i) $\mu^*L \sim_{\mathbb{Q}} A + E$ on $\tilde{X}$,
- (ii) $|\mathrm{vol}(A) - \mathrm{vol}(L)| < \varepsilon$, in particular $A^n > \mathrm{vol}(L) - \varepsilon$.

Theorem 2.2 (Fujita approximation (for big cohomology classes on compact Kähler manifold), [Bou02, Theorem 1.4]). *Let X be a compact Kähler manifold, and let $\alpha \in H^{1,1}(X, \mathbb{R})$ be a big class on X . Then, for every $\varepsilon > 0$, there exist a modification $\mu : \tilde{X} \rightarrow X$, a Kähler class ω , and an effective real divisor D on $\tilde{X}$ such that:*

- (i) $\mu^*\alpha = \omega + \{D\}$ as cohomology classes,
- (ii) $|\mathrm{vol}(\alpha) - \mathrm{vol}(\omega)| < \varepsilon$.

💡 **Idea of proof.**

3 Basic properties of volume functions

3.1 For $(\mathbb{R})$ -divisors

Volume depends only on the numerical class of an $\mathbb{R}$ -divisor.

Proposition 3.1 (Volume is numerical invariant, [FKL16, Theorem 3.5]). *Let D be an $\mathbb{R}$ -divisor on a proper normal variety X of dimension n . If D' is an $\mathbb{R}$ -divisor on X such that $D' - D$ is a numerically trivial $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor, then $\text{vol}(D) = \text{vol}(D')$.*

The volume function is continuous on $N^1(X)_{\mathbb{R}}$.

Proposition 3.2 (Continuity of volume function, [Laz04, Corollary 2.2.45]). *The function $\xi \mapsto \text{vol}(\xi)$ on $N^1(X)_{\mathbb{Q}}$ extends uniquely to a homogeneous continuous function*

$$\text{vol} : N^1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Volume increases in effective directions.

Proposition 3.3 (Volume increases in effective directions, [Laz04, Example 2.2.48]). *If $\xi \in N^1(X)_{\mathbb{R}}$ is big and $e \in N^1(X)_{\mathbb{R}}$ is effective, then*

$$\text{vol}(\xi) \leq \text{vol}(\xi + e).$$

The volume satisfies the birational invariance property.

Theorem 3.4. *Let*

$$v : X' \longrightarrow X$$

be a birational projective morphism of irreducible normal projective varieties. Then

$$\text{vol}_X(\xi) = \text{vol}_{X'}(v^*\xi)$$

for any class $\xi \in N^1(X)_{\mathbb{R}}$.

For a generically finite morphism we have a similar pullback formula.

Theorem 3.5. *Let $f : Y \rightarrow X$ be a proper, dominant, generically finite morphism of normal projective varieties over k . For any Cartier divisor D on X , we have*

$$\text{vol}_Y(f^*D) = \deg(f) \text{vol}_X(D).$$

Analogously to the case of global sections, the volume function satisfies the following property.

Theorem 3.6. *Let $g : X \rightarrow Y$ be a birational morphism of normal projective varieties, let D be an $\mathbb{R}$ -Cartier divisor on Y , and let G be an $\mathbb{R}$ -divisor on X . If $G - g^*D \geq 0$ is effective and g -exceptional, then*

$$\text{vol}(X, G) = \text{vol}(Y, D).$$

3.2 For cohomology classes

The volume function is continuous on the Bott–Chern cohomology space.

Proposition 3.7 ([Bou02, Corollary 4.11]). *The volume $\text{vol} : H^{1,1}(X, \mathbb{R}) \rightarrow \mathbb{R}$ is a continuous function.*

The volume function is log-concave.

Proposition 3.8 (Log concavity of volume function). *Let X be a compact Kähler manifold. If $\alpha, \alpha' \in H_{\text{BC}}^{1,1}(X, \mathbb{R})$ are big classes, then the volume satisfies the log concavity property*

$$\text{vol}(\alpha + \alpha')^{1/n} \geq \text{vol}(\alpha)^{1/n} + \text{vol}(\alpha')^{1/n}.$$

As for $\mathbb{R}$ -divisors, volume is birationally invariant for cohomology classes.

Proposition 3.9. *Let*

$$v : X' \longrightarrow X$$

be a birational morphism of irreducible compact Kähler manifolds. Then

$$\text{vol}_X(\xi) = \text{vol}_{X'}(v^* \xi)$$

for any class $\xi \in H_{\text{BC}}^{1,1}(X, \mathbb{R})$.

4 Upper semi-continuity of volumes

Jiao proved the following upper semi-continuity of volume for a projective family with reduced irreducible fibers.

Theorem 4.1 ([Jia25, Theorem 1.2]). *Let $f : X \rightarrow T$ be a projective flat morphism of varieties over $\mathbb{C}$ with reduced and irreducible fibers and D a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on X . Then for any $a \in \mathbb{R}$, the set*

$$\{t \in T \mid \text{vol}(D_t) < a\}$$

is open in T .

Boucksom proved the following upper semi-continuity of volume for smooth Kähler families.

Theorem 4.2 ([Bou02, Proposition 4.13]). *The volume is upper-semicontinuous in Kähler deformations in the following sense: if $\mathcal{X} \rightarrow S$ is a deformation of Kähler n -folds and $\alpha \in$*

$H^{1,1}(X, \mathbb{R})$ is a pseudoeffective class on the central fibre $X_0 =: X$, then one has

$$v(\alpha) \geq \limsup_{k \rightarrow +\infty} v(\alpha_k)$$

for every sequence $\alpha_k \in H^{1,1}(X_{t_k}, \mathbb{R})$ converging to α (with $t_k \rightarrow 0$).

Boucksom's proof contains two ingredients.

5 Deformation invariance of volume of adjoint divisors (classes)

The volume of log canonical divisors or adjoint classes satisfies deformation invariance properties (the proof will be given in a separate note).

Theorem 5.1 ([HMX18, Corollary 4.3]). *Let $\pi : X \rightarrow T$ be a projective morphism of smooth varieties. Suppose that (X, Δ) is log canonical and has simple normal crossings over T . Then the volume function*

$$t \mapsto \text{vol}(X_t, K_{X_t} + \Delta_t)$$

is independent of t .

This is also true for generalized Kähler pairs, when the central fiber is projective with a big adjoint class.

Theorem 5.2 (HLR25'). *Let $f : X \rightarrow S$ be a smooth family from a Kähler manifold X onto a smooth, connected, relatively compact curve S . Fix a point $0 \in S$ and assume that $(X, B + \beta)$ is a generalized klt pair such that (X, B) is log smooth over S and $\beta = \bar{\beta}$ where β is nef over S and X_0 is projective. If $K_{X_0} + B_0 + \beta_0$ is big, then the volume function*

$$t \mapsto \text{vol}(K_{X_t} + B_t + \beta_t)$$

is constant on some neighborhood $U \subset S$ of 0 .

6 Volume in divisorial Zariski decomposition

We have already seen that the volume may increase in the effective direction; in this section we show that subtracting the negative part of the Zariski decomposition does not change the volume.

7 Volume in the minimal model program

8 Jiao's proof of upper semi-continuity of volume function

Theorem 8.1. *Let $X \rightarrow T$ be a projective flat morphism of varieties and D a Cartier divisor on X . Suppose X_η is the generic fiber of f .*

Write $D_\eta := D|_{X_\eta}$ and $D_t := D|_{X_t}$ for every closed point $t \in T$, then

$$\mathrm{vol}(D_\eta) = \inf_{t \in T'} \mathrm{vol}(D_t)$$

for any dense subset $T' \subset T$.

💡 Idea of proof. We divide the proof into several steps.

Step 1. We first reduce the problem to showing that when D_η is big, for any $\epsilon > 0$ there exists some $t \in T'$ such that

$$\mathrm{vol}(D_t) < \mathrm{vol}(D_\eta) + \epsilon.$$

Step 2. We then try to apply the Fujita approximation on the generic fiber; however, the problem is that the generic fiber is not defined over an algebraically closed field (which is needed for the Fujita approximation). Thus we first pass to the algebraic closure of the function field $\bar{K}$, and we get $g_\eta^* D_\eta \sim_{\mathbb{Q}} A_\eta + E_\eta$, where Y_η is smooth, $E_\eta \geq 0$, A_η is ample, and

$$\mathrm{vol}(D_\eta) \leq \mathrm{vol}(A_\eta) + \epsilon,$$

and then descend to the generic fiber X_η . We then extend the Fujita approximation to the smooth compactification Y , as illustrated by the following diagram:

$$\begin{array}{ccccccccc} Y & \longleftarrow & Y_\eta & \longrightarrow & X_\eta & \longrightarrow & X_\eta & \longrightarrow & X \\ \downarrow & & \searrow & & \downarrow & & \downarrow & & \downarrow \\ T & \longleftarrow & \mathrm{Spec}(\bar{K}) & \longrightarrow & \mathrm{Spec}(\eta) & \longrightarrow & T & & \end{array}$$

where

$$g^* D \sim_{\mathbb{Q}} A + E,$$

with A ample over T . If

$$g_t^* D_t \sim_{\mathbb{Q}} A_t + E_t$$

is the Fujita approximation on the fiber, then we are done, since

$$\mathrm{vol}(D_\eta) \geq A_\eta^d = \mathrm{vol}(A_\eta) \geq \mathrm{vol}(D_\eta) - \epsilon$$

and

$$\mathrm{vol}(D_t) \geq A_t^d = \mathrm{vol}(A_t) \geq \mathrm{vol}(D_t) - \epsilon.$$

Thus we are done, since $A_t^d = A_\eta^d$ by the constancy of intersection numbers in flat families.

Step 3. In general, we need to take a Fujita approximation of $g_t^*D_t$, and get that for any $\epsilon' \in \mathbb{R}_+$ there exists a projective birational morphism $h_t : W_t \rightarrow Y_t$ such that

$$h_t^*g_t^*D_t \sim_{\mathbb{Q}} h_t^*(A_t + E_t) \sim_{\mathbb{Q}} A'_t + E'_t.$$

We then apply the asymptotic orthogonality estimate and the Hodge-type inequality:

$$A_\eta^{d-1} \cdot (A_\eta + E_\eta) \leq A_\eta^d + C'\sqrt{\epsilon}$$

and

$$\left(A_t^d\right)^{\frac{d-1}{d}} \left(A_t'^d\right)^{\frac{1}{d}} \leq (h_t^*A_t)^{d-1} \cdot A'_t$$

to show that $A_t'^d$ is not far from A_t^d :

$$A_t'^d \leq \left(\left(A_t^d\right)^{\frac{1}{d}} + \frac{C'\sqrt{\epsilon}}{\left(A_t^d\right)^{\frac{d-1}{d}}} \right)^d.$$

Thus $g_t^*D_t \sim_{\mathbb{Q}} A_t + E_t$ already serves as a good substitute for the Fujita approximation.

References

- [Bou02] S. Boucksom, *On the volume of a line bundle*, International Journal of Mathematics **13** (2002), no. 10, 1043–1063.
- [FKL16] M. Fulger, J. Kollár, and Brian Lehmann, *Volume and Hilbert function of $\mathbb{R}$ -divisors*, Michigan Math. J. **65** (2016), no. 2, 371–387.
- [HMX18] Christopher D. Hacon, James McKernan, and Chenyang Xu, *Boundedness of moduli of varieties of general type*, J. Eur. Math. Soc. (JEMS) **20** (2018), no. 4, 865–901.
- [Jia25] Junpeng Jiao, *The volume function is upper semicontinuous on families of divisors*, 2025.
- [Laz04] Robert Lazarsfeld, *Positivity in algebraic geometry. I*, vol. 48, Springer-Verlag, Berlin, 2004, Classical setting: line bundles and linear series.